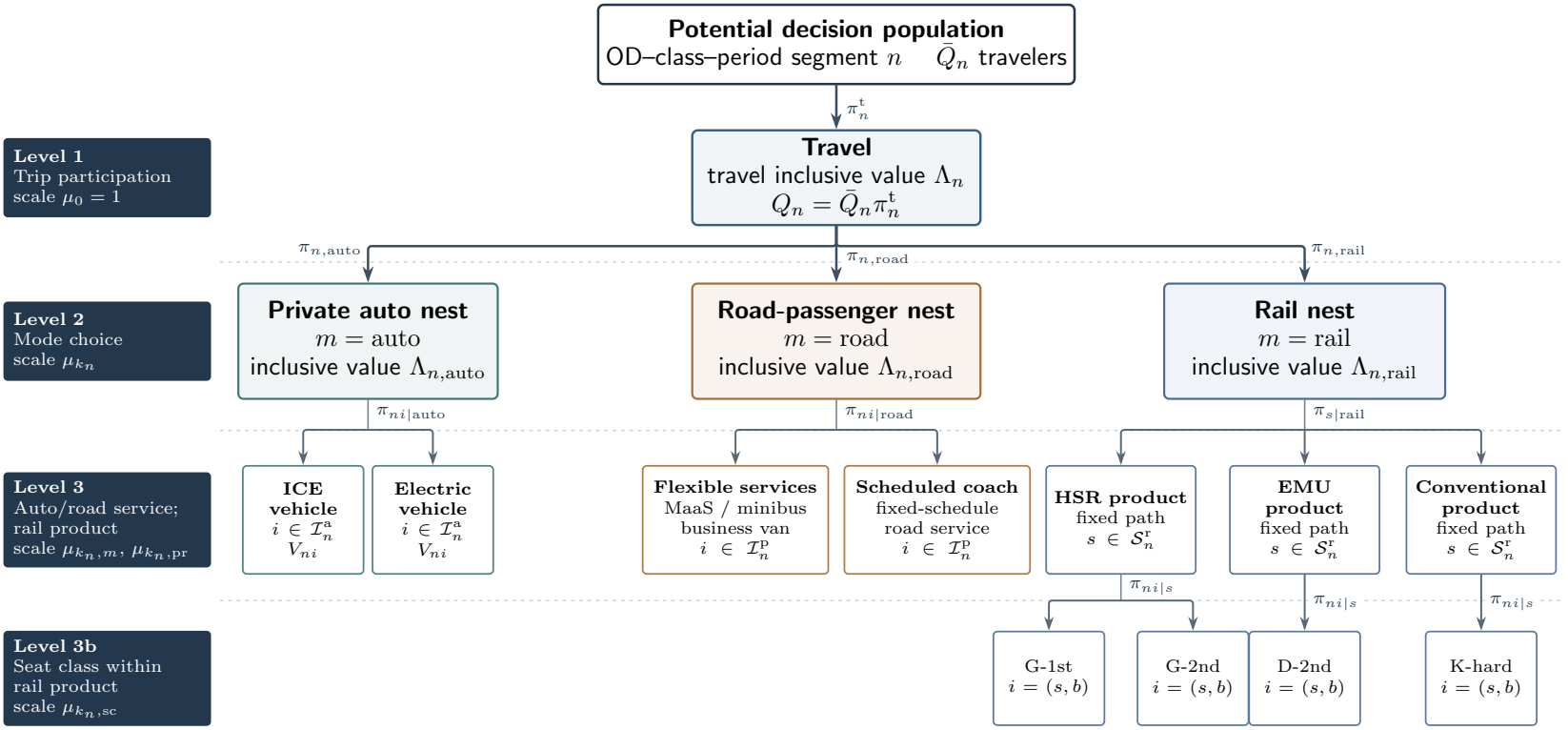
Three-level elastic nested-logit demand for segment $n = (w, k, h)$

Given railway design z and economic state ξ: available sets $\mathcal{I}_{nm}(z)$ and utilities $V_{ni}(z, \xi)$



Detailed realized flow $q_{ni} = \underbrace{\bar{Q}_n}_{\text{potential}} \underbrace{\pi^{\mathrm{t}}_n}_{\text{participation}} \underbrace{\pi_{n,m(i)}}_{\text{mode}} \underbrace{\pi_{ni|m(i)}}_{\text{detailed}}$

For rail, $\pi_{ni|m(i)} = \pi_{s(i)|\mathrm{rail}}\, \pi_{ni|s(i)}$.

No-travel flow $q^0_n = \bar{Q}_n(1 - \pi^{\mathrm{t}}_n)$ and $q^0_n + \sum_i q_{ni} = \bar{Q}_n$

Random-utility consistency: $0 < \mu_{k_n,\mathrm{sc}} \le \mu_{k_n,\mathrm{pr}} \le \mu_{k_n} \le \mu_0 = 1$ and $0 < \mu_{k_n,m} \le \mu_{k_n}$ for auto and road; an empty nest is omitted from $\mathcal{M}_n(z)$.